Ranking Flaky Test Failures

Flaky tests pass and fail on the same code without any change to the program under test. This study runs each test of 26 open-source Java projects one hundred times and records every outcome together with the order in which the tests ran.

The detector labels 450 of 533 tests as stable, 51 as flaky, and 32 as unclear. A manual check of the flaky tests confirms 47 of the 51 labels, and the remaining four tests depend on the system clock.